CHEMISTRY | ADAPTLY

SEPARATING MIXTURES – FILTRATION & DISTILLATION

KEY VOCABULARY

Filtration: separates an insoluble solid from a liquid.

Crystallisation: separates a soluble solid from its solution.

Distillation: separates a liquid from a solution by evaporating and condensing.

Chromatography: separates substances by how far they travel in a solvent.

COMMON MISCONCEPTIONS

~~Filtration separates salt from water.~~

✓ Salt dissolves — use crystallisation or distillation.

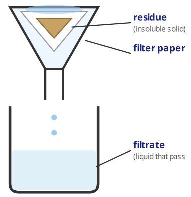
~~The residue is the liquid that passes through.~~

✓ The residue is the solid left in the filter paper.

~~Distillation makes water disappear.~~

✓ The vapour is cooled and condensed back to liquid.

FILTRATION



1 CHOOSE A METHOD

Name the best method to separate sand from water. **[1 mark]**

2 RESIDUE & FILTRATE

In filtration, define the terms **residue** and **filtrate**. **[2 marks]**

3 GET PURE WATER

Explain how distillation produces pure water from salty water. **[3 marks]**

4 RECOVER A SALT

Name the method used to recover copper sulfate crystals from solution. **[1 mark]**

5 FILL THE GAPS

Complete the sentence using the word bank. **[2 marks]**

Filtration removes an _____ solid. In distillation the liquid _____ then _____ on a cold surface.

insoluble | soluble | evaporates | condenses

6 CHROMATOGRAPHY

Explain why some inks travel further up the paper than others. **[2 marks]**

TIP: *Dissolved (soluble) solids pass through filter paper — only insoluble solids are trapped.*

✓ Think: Why must the thermometer sit at the top of the flask in distillation?